



Topic: Conceptual Modeling and Logical Modeling

A Case Study

A digital marketing company wants to develop a database to support the management of its clients' social media accounts. The company works with various brands and is responsible for planning, creating, and publishing content on different social media platforms. Each customer has a record in the system where some basic information is stored, such as the name of the company, the address of the headquarters, the sector of activity, their contacts and the official website, if any. A client may have multiple social media accounts, for example on platforms such as Instagram, Facebook, or TikTok. For each account, data such as the platform used, the username and the date of account creation are recorded. The company has several social media managers, responsible for managing the accounts. It should be saved when a manager started/finished managing a certain account, as well as the total amount they received. On the different accounts, several publications are made, which correspond to the content published on social networks. For each publication, information such as the date and time of publication, the type of content (image, video or text), a brief description of the content, as well as the comments recorded with the text of the comment and the date and time it was made, are recorded. Some comments may also appear as a response to other comments on a post.

Based on the case study presented and taking into account the various steps of the database development methodology, it is intended that:

- a) Develop a list of description requirements, which allows you to develop a conceptual schema for a database for the case presented.
- b) Using the Chen notation, sketch a conceptual scheme for a database that is capable of hosting the information revealed by the case presented.
- c) Convert the developed conceptual schema to your possible logical schema. List and justify each of the steps taken in the conversion process.